DMI COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MimicModel Securing Cloud Data File Sharing with Homomorphic Encryption and Blockchain for Confidentiality and Integrity

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OBJECTIVE

- **Implement Homomorphic Encryption** to ensure **data confidentiality** during storage and processing.
- Use Blockchain Technology to maintain **tamper-proof records** of data integrity.
- Facilitate Transparent & Secure Data Sharing using **smart contracts** for automated access control.
- Enable Real-Time Verification of unauthorized modifications through **periodic hash verification**.
- Enhance Trust & Security by eliminating reliance on a single authority and ensuring a decentralized, auditable system.

ABSTRACT

- Cloud Storage Security Risks – Cloud storage is widely used but faces threats like data leakage, mismanagement, and malicious attacks, leading to loss of user control over data.
- Proposed Security Model – Integrates cloud computing, blockchain technology, and **homomorphic encryption to ensure data confidentiality and integrity.**
- Blockchain-Based Data Integrity – The **Cloud Service Provider (CSP)** computes a master hash value of its database and stores it on a private blockchain, allowing clients to verify data integrity.
- Secure Data Sharing via Smart Contracts – **Smart contracts** regulate data access and sharing, ensuring transparent and secure transactions with escrow mechanisms to prevent fraud.
- Comprehensive Security Properties – The model guarantees confidentiality, integrity, privacy, non-repudiation, and anonymity, making cloud storage secure and reliable.

LITERATURE SURVEY

SNO	TITLE	AUTHOR	DESCRIPTION	MERITS	DEMERITS
1	A Blockchain-Based Secure Data Sharing Framework for Edge Computing	Xiaoyu Zhu et al	This paper proposes a blockchain-based secure data sharing framework for edge computing, which enables secure data sharing among edge nodes.	Provides a secure and decentralized data sharing solution, ensures data integrity and authenticity.	May require high computational overhead for encryption and decryption.
2	Secure Data Sharing Among Edge Nodes Using Blockchain and IPFS	Zehui Xiong et al	This paper proposes a secure data sharing framework among edge nodes using blockchain and InterPlanetary File System (IPFS), which ensures secure and decentralized data sharing.	Provides a decentralized storage solution, ensures data integrity and authenticity.	May require complex smart contract design and management.
3	Trust Management for Artificial Intelligence: A Standardization Perspective	Tai-Won Um, Jinsul Kim, Sunhwan Lim, Gyu Myoung Lee	The paper proposes a trust management framework for AI systems, emphasizing standardization. It covers trust requirements across data, models, and application layers, focusing on explainability, data quality, fairness, and privacy.	Emphasis on ethical AI and global standards. Comprehensive trust indicators like explainability and robustness.	High implementation complexity. Limited real-world case studies. Resource-intensive monitoring and validation.

SNO	TITLE	AUTHOR	DISCRIPTION	MERITS	DEMERITS
4	A Secure Data Sharing Framework for Edge Computing Using Blockchain and Attribute-Based Encryption	Liang Zhao et al.	This paper proposes a secure data sharing framework for edge computing using blockchain and attribute-based encryption, which ensures secure and decentralized data sharing.	This paper proposes a secure data sharing framework for edge computing using blockchain and attribute-based encryption, which ensures secure and decentralized data sharing.	May require complex smart contract design and management. .
5	A Data Encryption and File Sharing Framework Among Microservices-Based Edge Nodes with Blockchain	Weimin Li, ZiTong Li, Zhengmao Yan, Yi Liu, Detian Zeng, Haoyang Yu, Wenxiong Chen, and Fan Wu	The framework combines Ethereum blockchain, IPFS, and HDFS to manage and share encrypted files among edge nodes. Multi-round AES encryption enhances data security, while zero-knowledge proofs (ZKP) verify node identities.	Robust data security using multi-round AES encryption. Decentralized and tamper-resistant file sharing with blockchain. Effective identity verification with ZKP.	Increased computational load due to multi-round encryption and ZKP. Complexity in managing decentralized edge environments and communication
6	Research on Distributed Secure Storage Framework of Industrial Internet of Things Data Based on Blockchain	Hongliang Tian, Guangtao Huang	This study proposes a two-tier distributed secure storage framework for Industrial Internet of Things (IIoT) data, combining edge computing and blockchain technology. The framework consists of an edge network layer for preprocessing industrial data and a blockchain storage layer for secure and traceable data storage.	Enhances security and traceability of IIoT data through blockchain. Provides a scalable storage solution by combining on-chain and off-chain storage methods.	Complex implementation due to the integration of multiple technologies. Potential challenges in managing data consistency between edge and blockchain layers.

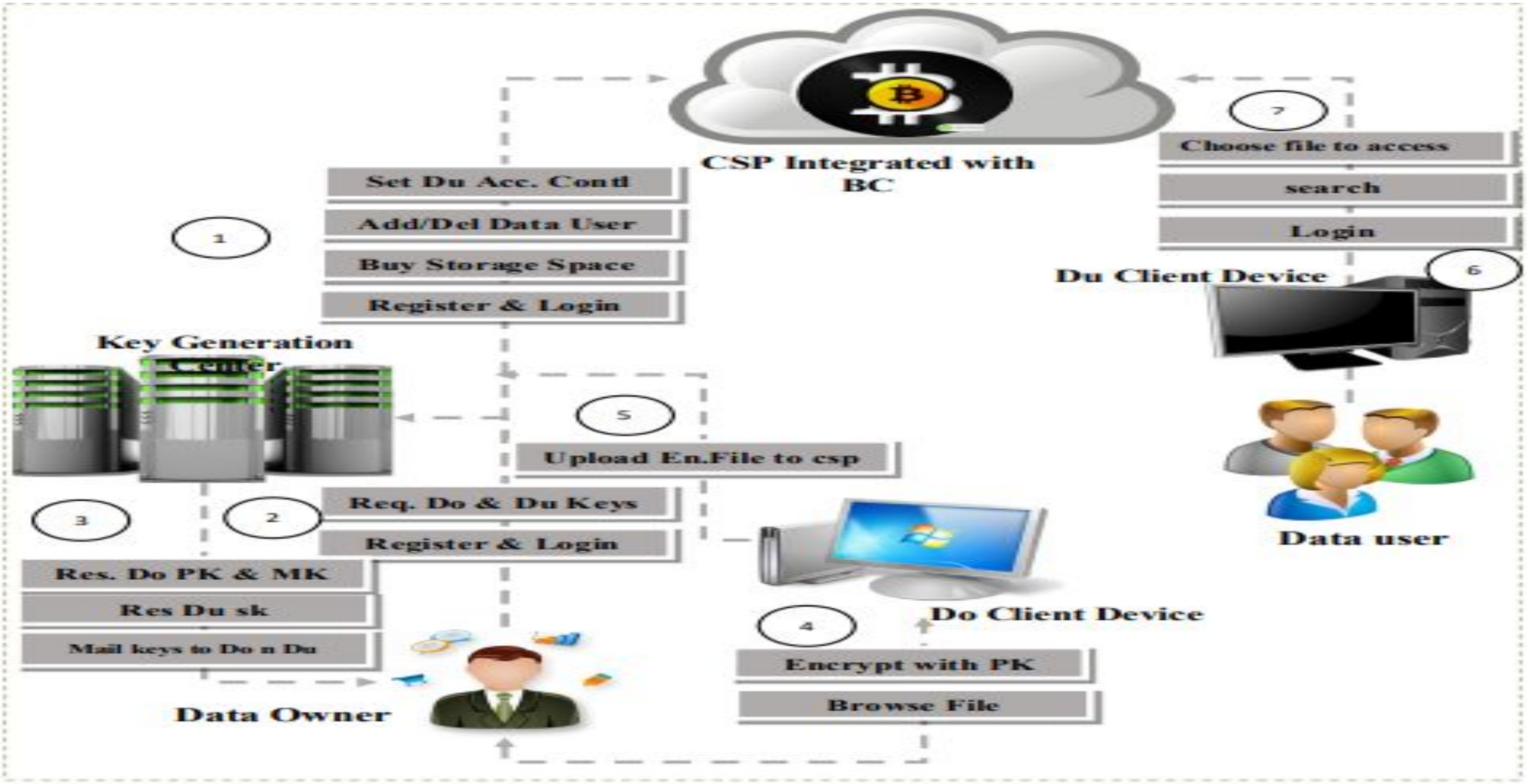
EXISTING SYSTEM

- Traditional cloud storage security relies on centralized encryption and authentication.
- **Data is encrypted using AES/RSA**, but decryption is required for processing, exposing it to threats.
- Access control mechanisms rely on **passwords, OAuth, or ACLs**, which can be vulnerable to attacks.
- Audit logs are centralized and can be tampered with, reducing reliability.
- Unauthorized modifications are not detected immediately, leading to potential security breaches.
- Cloud providers have full control over user data, creating trust issues.
- Weak authentication mechanisms increase the risk of unauthorized access.
- Logs and access records are stored centrally, making them prone to modification.

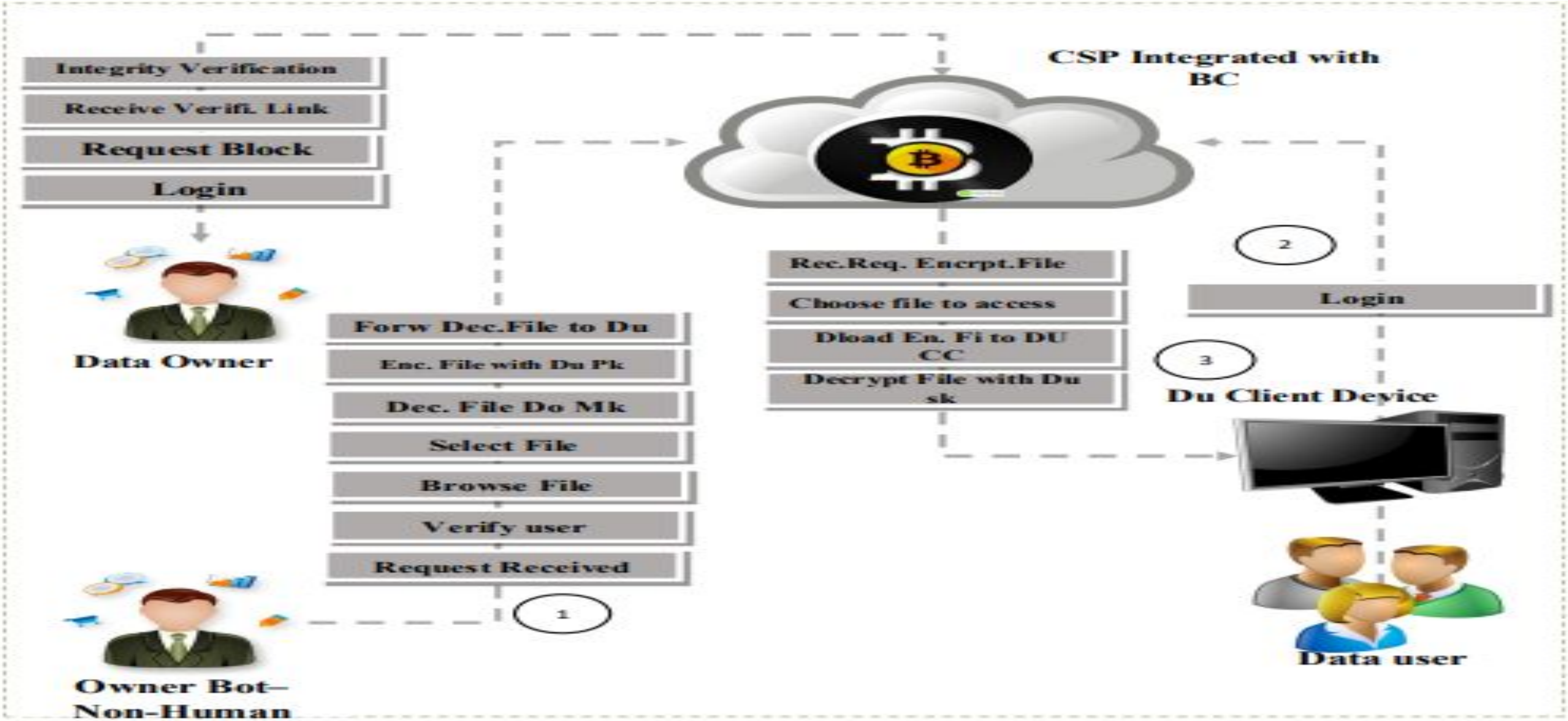
PROPOSAL SYSTEM

- **Integrates Homomorphic Encryption:** Allows data processing while encrypted, ensuring confidentiality without the need for decryption.
- **Blockchain for Data Integrity:** Stores hash values of encrypted data on a blockchain, making any tampering detectable in real-time.
- **Periodic Hash Verification:** Cloud service providers generate hash values periodically and store them on the blockchain for real-time integrity verification.
- **Tamper-Proof Audit Trail:** Blockchain ensures immutable records, preventing unauthorized modifications.
- **Enhanced Security & Trust:** Users can verify who accessed or modified data, reducing security risks.
- **Minimized Single Authority Dependence:** Decentralized verification ensures no single entity controls or alters security measures.

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE



CONCLUSION

- The project provides a robust and secure solution for managing sensitive data in the cloud by integrating homomorphic encryption and blockchain technology.
- This system ensures data confidentiality through encryption, while blockchain offers a tamper-proof mechanism for verifying data integrity.
- The model's ability to handle secure file sharing using public-private key encryption and smart contracts enhances trust and transparency between data owners, users, and cloud service providers. Additionally, its audit capabilities empower data owners to monitor and verify all transactions.
- Mimic Bot addresses modern cloud security challenges effectively, offering a scalable, reliable, and efficient framework for safeguarding sensitive information in a rapidly digitalizing world.

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THANK YOU